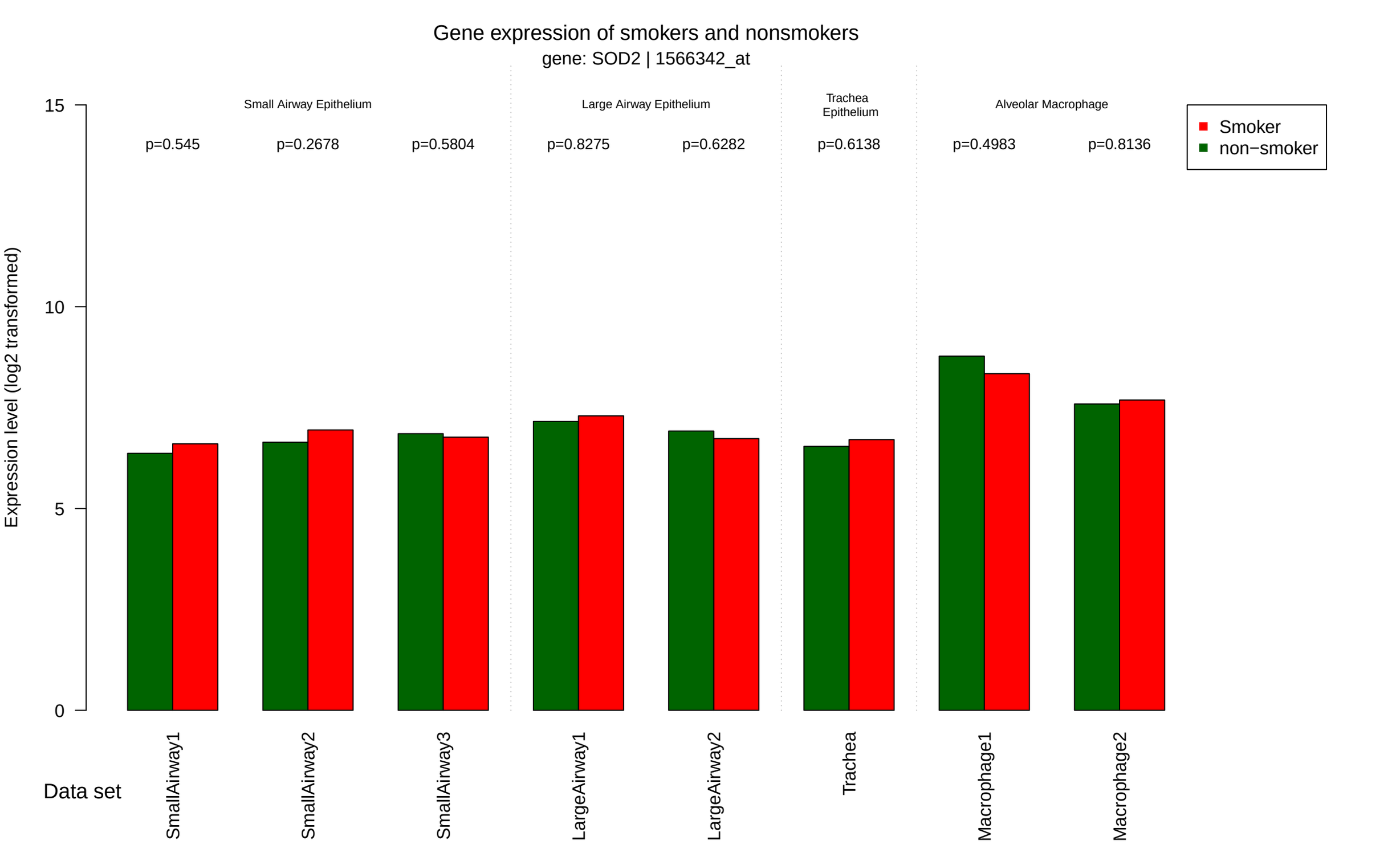
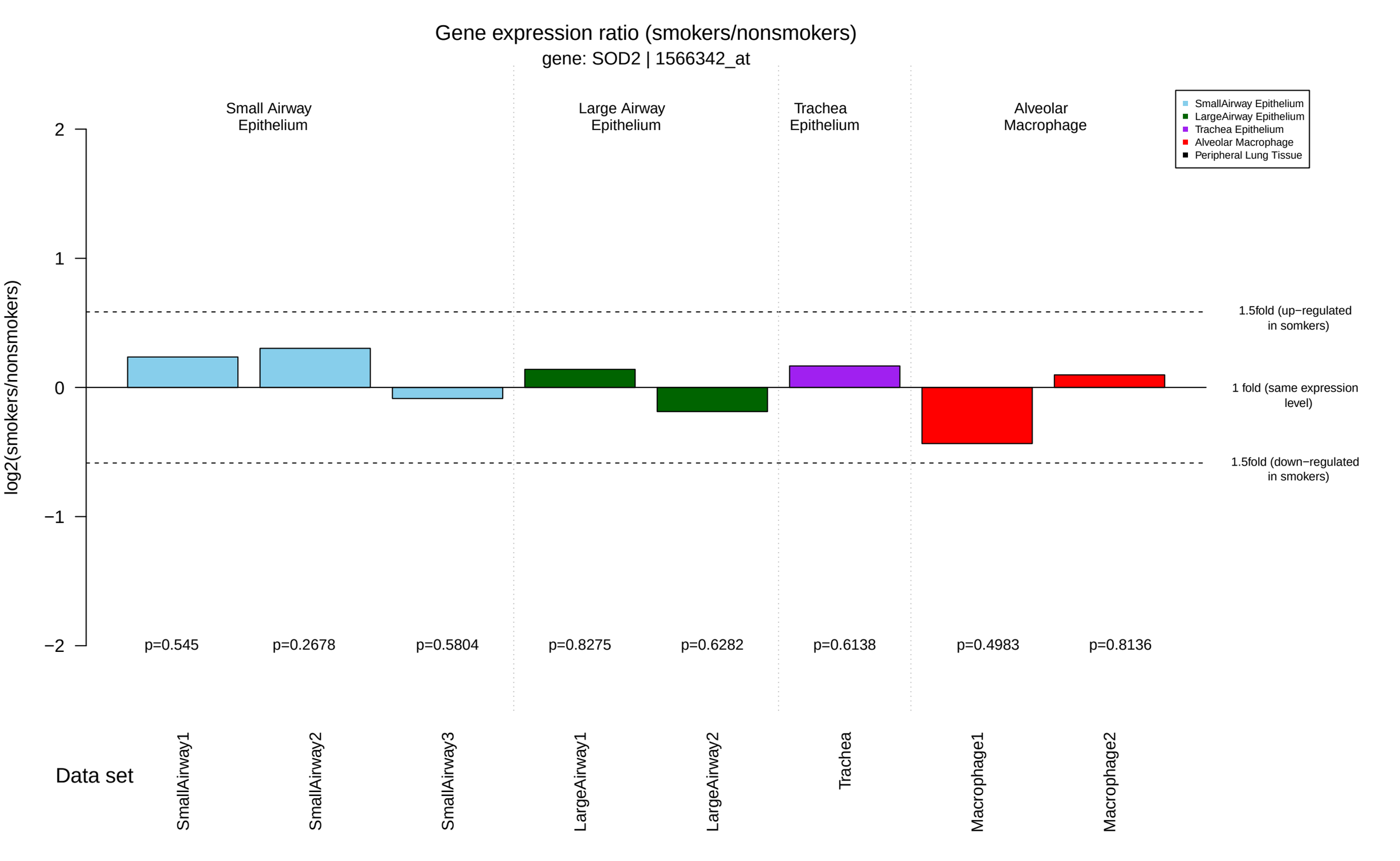
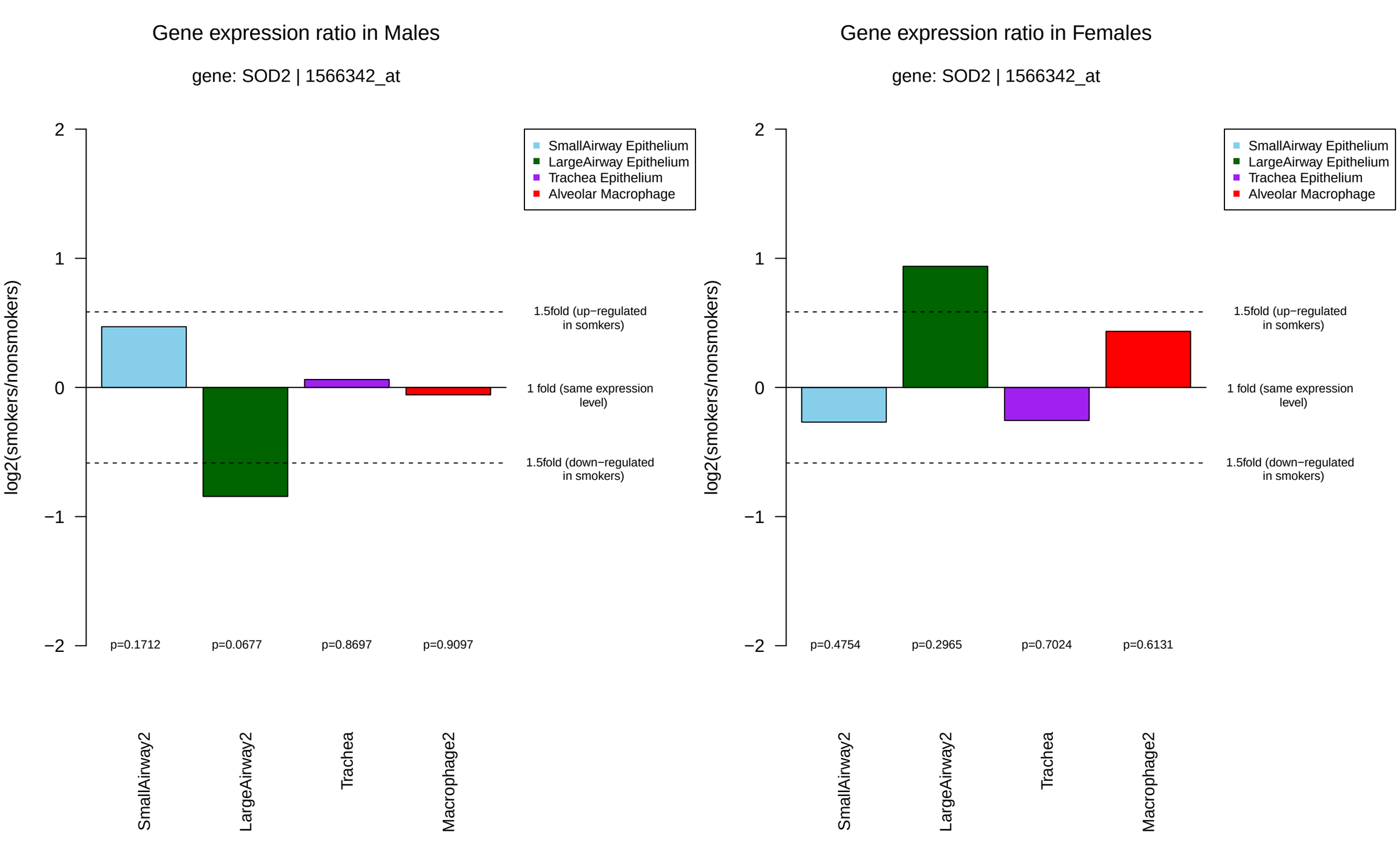
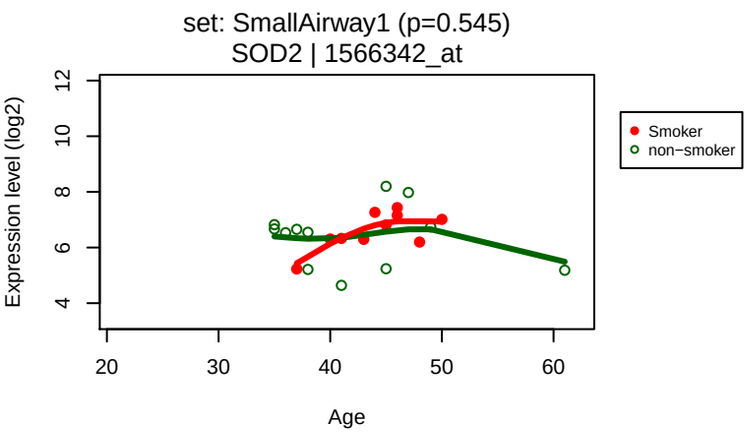
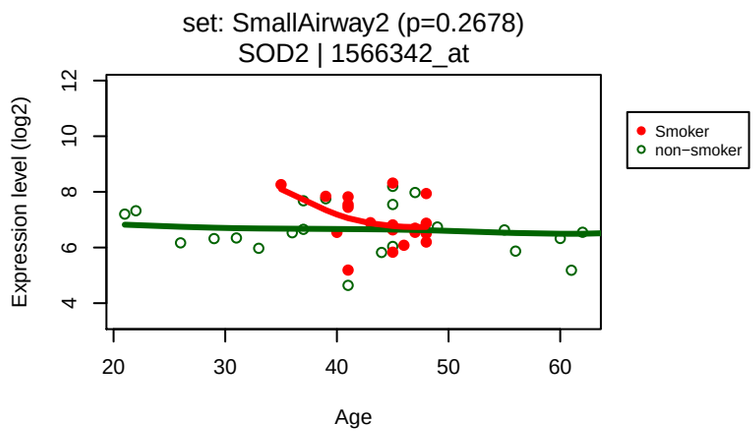
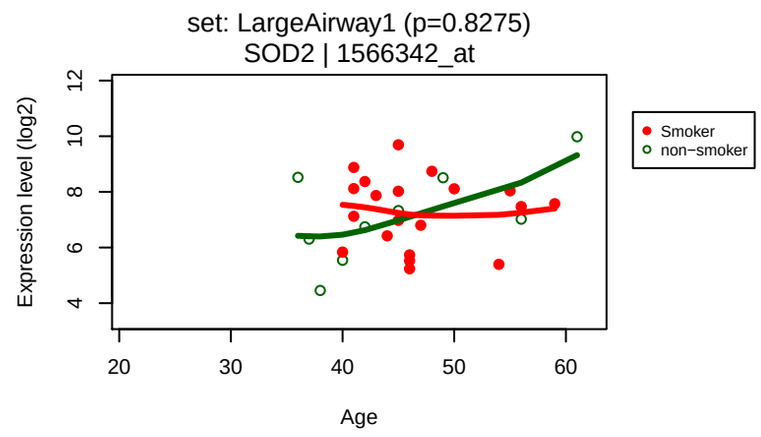

Gene expression of smokers and nonsmokers
gene: SOD2 | 1566342_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.545
p=0.2678
p=0.5804
p=0.8275
p=0.6282
p=0.6138
p=0.4983
p=0.8136
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

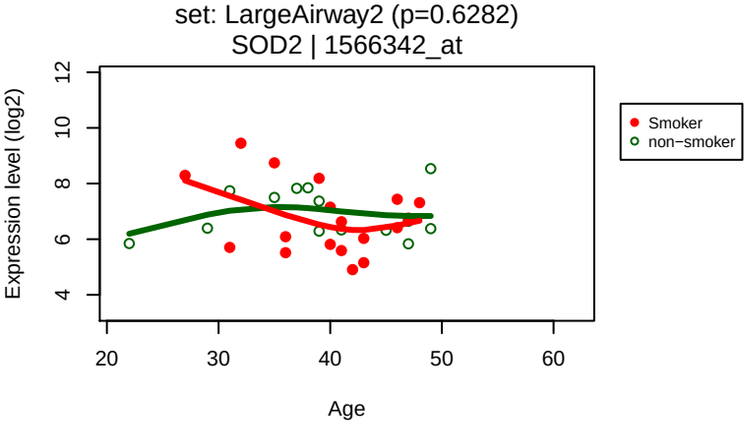

Gene expression ratio (smokers/nonsmokers)
gene: SOD2 | 1566342_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.545
p=0.2678
p=0.5804
p=0.8275
p=0.6282
p=0.6138
p=0.4983
p=0.8136
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

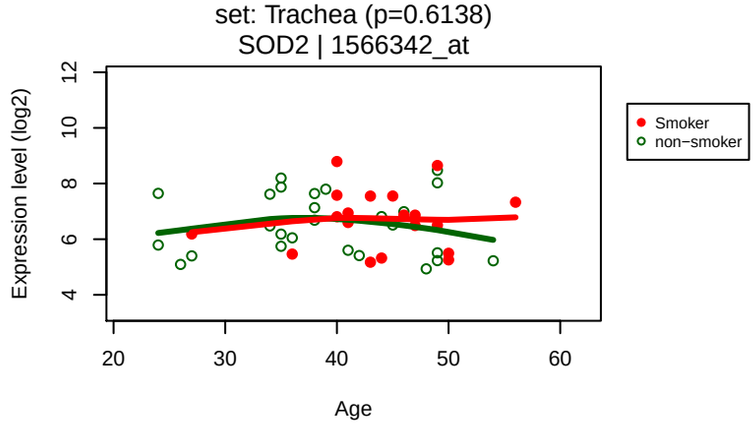

Gene expression ratio in Males
gene: SOD2 | 1566342_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1712
p=0.0677
p=0.8697
p=0.9097
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOD2 | 1566342_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4754
p=0.2965
p=0.7024
p=0.6131
SmallAirway2
LargeAirway2
Trachea
Macrophage2

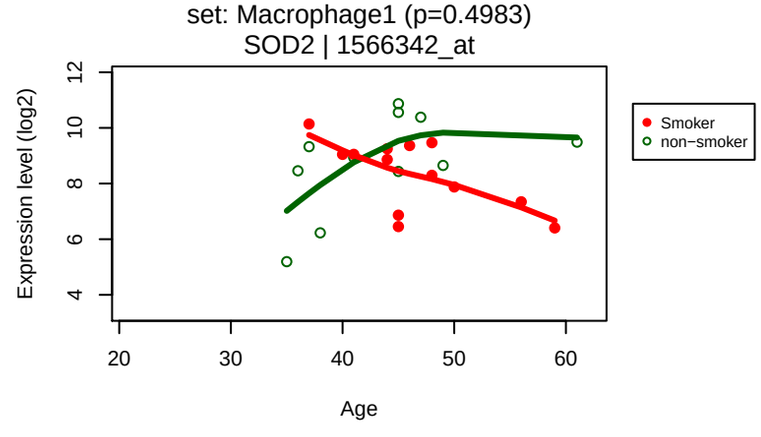

set: SmallAirway1 (p=0.545)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker

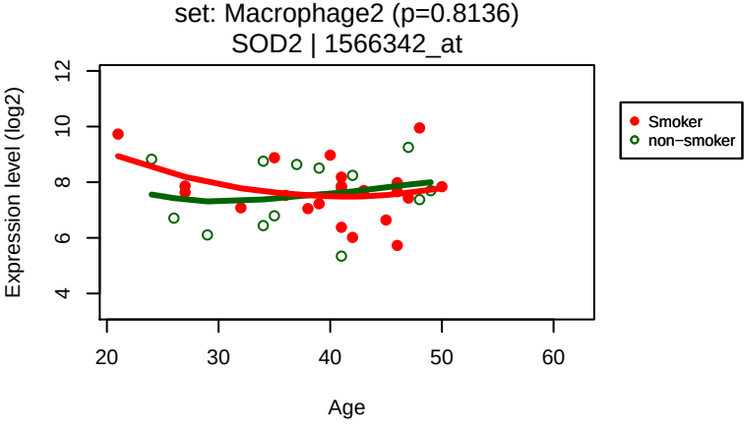

set: SmallAirway2 (p=0.2678)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker

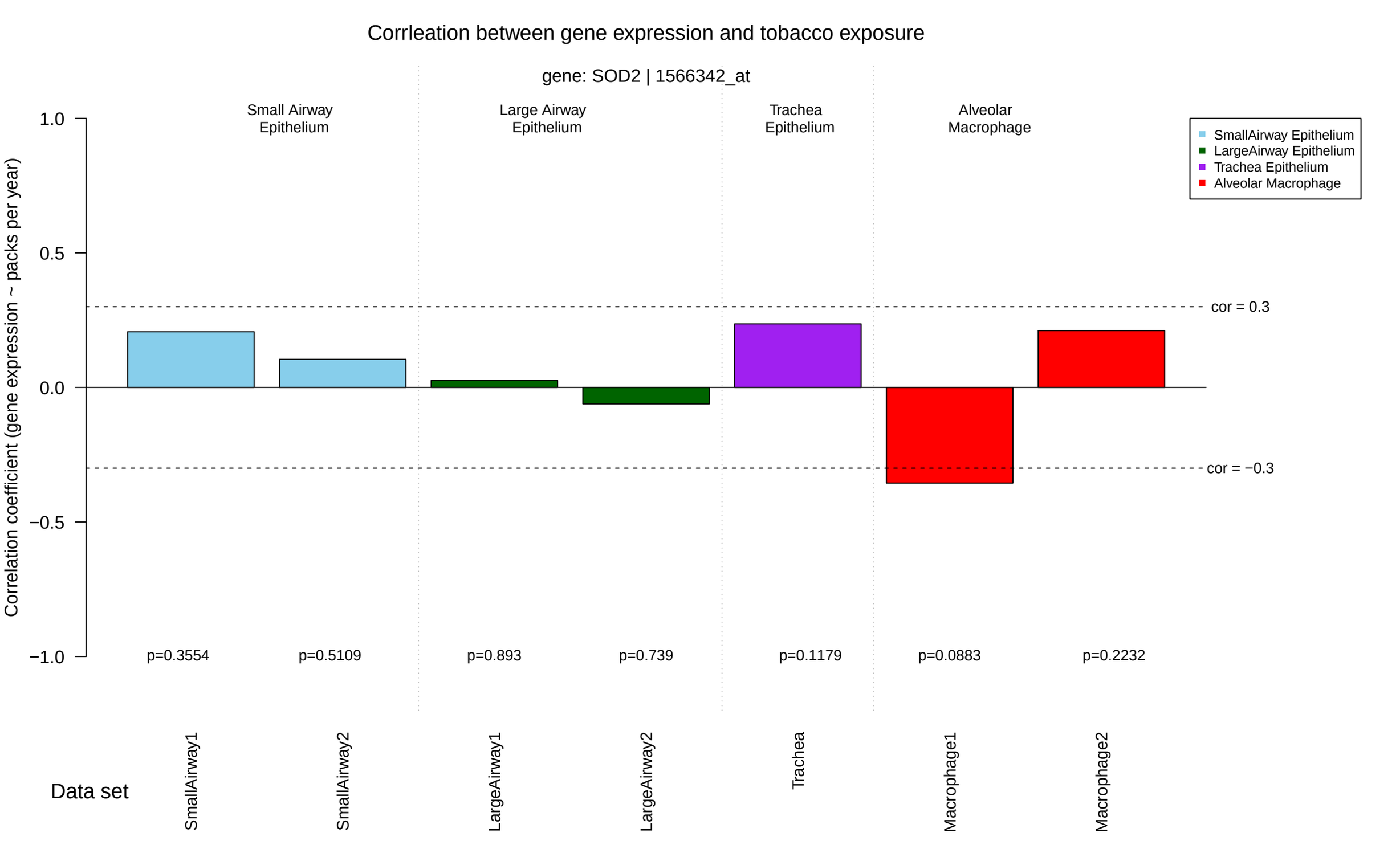

set: LargeAirway1 (p=0.8275)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6282)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6138)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4983)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8136)
SOD2 | 1566342_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOD2 | 1566342_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3554
p=0.5109
p=0.893
p=0.739
p=0.1179
p=0.0883
p=0.2232
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2